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Studierichting: Informatica
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Opgave: $z^3 = 1 - i$

$$r = \sqrt{1^2 + (-1)^2} = \sqrt{1} = \sqrt{2}$$

$$\tan \theta = \frac{-1}{1} = -1 \Rightarrow \theta = \text{Bgtan}(-1) = -\frac{\pi}{4}$$

$$\Rightarrow w = \sqrt[3]{2} \operatorname{cis} \left(\frac{-\frac{\pi}{4} + k \cdot 2 \cdot \pi}{3} \right)$$

$$w_0 = \sqrt[3]{2} \cdot \operatorname{cis} \left(\frac{-\frac{\pi}{4}}{3} \right) = \sqrt[3]{2} \operatorname{cis} \cdot \left(-\frac{\pi}{12} \right)$$

$$w_1 = \sqrt[3]{2} \cdot \operatorname{cis} \left(\frac{-\frac{\pi}{4} + 2\pi}{3} \right) = \sqrt[3]{2} \cdot \operatorname{cis} \left(\frac{7\pi}{12} \right)$$

$$w_2 = \sqrt[3]{2} \cdot \operatorname{cis} \left(\frac{-\frac{\pi}{4} + 4\pi}{3} \right) = \sqrt[3]{2} \cdot \operatorname{cis} \left(\frac{5\pi}{4} \right) = -\sqrt[3]{2} \cdot \frac{\sqrt{2}}{2} \cdot (1 + i)$$

References